

RAHUL A GOWDA

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PROFESSIONAL SUMMARY

AI Engineer experienced in building **Generative and Agentic AI systems**, developing **Scalable multi-agent systems**, **RAG pipelines**, and **end-to-end AI tools** using LangChain, LangGraph, and **LLMs** with **fine-tuning** and **transfer learning**. Skilled in **STT**, **TTS**, and **multi-modal models**, as well as **ML/DL/Neural Network** development for predictive analytics. **Worked on advanced AI projects at ISRO (URSC), delivering production-ready solutions that bridge intelligent reasoning with practical deployment.**

TECHNICAL SKILLS

Programming Languages: Python, Java, C

Tools & Frameworks: Scikit-Learn, TensorFlow, Hugging Face, OpenCV, PyTorch, Docker, FastAPI, Flask, Jira

Machine Learning & Deep Learning: Model Development, Neural Networks, NLP, Deployment

Generative and Agentic AI: Finetuning LLM, LangChain, LangGraph, crewAI, Phidata, LangSmith, NLP, Vector Databases, Graph Databases, RAG, Azure AI foundry, LightRAG, Unsloth, BitsandBytes, Azure cloud, MCP Servers

PROFESSIONAL EXPERIENCE

Webknot technologies Pvt. Ltd., Bangalore, India: Senior AIML Engineer Dec. 2025 - present

- Owned end-to-end development of **PrepMeToWork**, an AI-powered mock interview and assessment platform for students and institutions.
- Architected long-term context and memory systems using vector databases for adaptive interview experiences.
- Built real-time multi-agent pipelines with LangGraph and LiveKit for interview generation, evaluation, and feedback.
- Led **Alnization**, designing multi-agent systems to automate organizational workflows.
- Developed HR automation agents for candidate screening, interview orchestration, and hiring support using LangGraph.
- Optimized stateful agent workflows to reduce human effort while maintaining scalability and reliability.

iTCart Pvt. Ltd., Bangalore, India: AI Executive – Founder's Office Jun. 2025 – Dec. 2025

- Engineered a full-stack AI solution for a LIS company's LinkedIn automation, leveraging LangChain, LangGraph, and MCP Servers to streamline posting, lead generation, and messaging workflows.
- Led AI R&D initiatives under the Founder's Office, designing and deploying GenAI tools using frameworks such as LangChain, LangGraph, and LangSmith.
- Collaborated with USL for AiXRush (Real-time Unified Sports Hub) to architect and automate the Proof of Concept (PoC) for seamless operations.
- Proposed, architected, and guided the integration of AI-driven capabilities into AiXFieldGo, AiXCollect Pro, and AiXQP, enhancing automation and data intelligence.
- Built and evaluated multi-agent systems utilizing LangGraph, LangChain, and OpenAI LLMs, ensuring performance optimization through LangSmith-based analysis.

URSC (ISRO), Bangalore, India: Project Trainee Mar. 2025 – Jun. 2025

- Engineered a suite of Generative and Agentic AI applications (Gyaan Meet, Docx, Coder, Admin) with Team Gyaan, enabling autonomous document summarization, coding assistance, and meeting insights.
- Researched LLM intelligence by architecting a novel evaluation framework that enabled smaller LLMs to tackle complex questions through the decomposition provided by a larger model.
- Developed a CNN-based bone fracture detection model with SHAP-based explainability, providing transparent and interpretable outputs for medical review.
- Formulated methods for evaluating the intelligence of LLMs.
- Created LightRAG-powered chatbots to enable quick knowledge retrieval from technical documents, reducing search time for scientists by simultaneously adding new information to the graph database for both short-term and long-term knowledge persistence.

AI-Zira Technologies, Remote: AI Engineer Intern (Intern Lead)

Oct. 2024 – Mar. 2025

- Led a team of 10 interns to build production-ready agentic AI tools using LangChain and LangGraph, including an AI Email Writer, a Medical Assistant chatbot, and eight more applications.
- Integrated Neo4j knowledge graphs, vector embeddings, and email workflows into Dockerized AI chatbots, enabling context-aware responses, persistent knowledge storage, and streaming capabilities.
- Coordinated code reviews, deployment pipelines, and model performance optimization for multi-agent workflows.

Aspire Technologies, Bangalore, India: AI Business Intelligence Intern

Sept. 2024 – Mar. 2025

- Designed and implemented preprocessing pipelines for large-scale structured datasets, ensuring data consistency and quality.
- Trend analysis was performed using Pandas, NumPy, Scikit-learn, and Seaborn to provide actionable insights for decision-making.
- Developed and deployed machine learning and deep learning inference pipelines that reduced model execution time.

Unifirst Robotics, Bangalore, India: Gen AI Engineer Intern

Oct. 2023 – Nov. 2023

- Built and optimized ML/DL models for predictive analytics, focusing on improving both interpretability and performance.
- Performed transfer learning with CNN models to significantly improve classification and detection accuracy in visual inspection tasks.
- Developed sentiment analysis and object detection systems achieving high accuracy and robust performance
- Explored Generative AI concepts, particularly transformer architectures, to understand their mechanisms and potential in enhancing NLP and vision-based AI solutions.

PROJECTS

SmartHealth Assistant – AI Medical Chatbot with Graph Database

2025

Built a production-ready LangGraph system for symptom and vital-based medical assistance.

- Built a LangGraph system to analyze symptoms/vitals and generate insights via Google Gemini.
- Modeled patient data as graphs in Neo4j for efficient retrieval and longitudinal tracking.
- Dockerized and deployed the end-to-end application for scalable production use.
- Tech stack: LangGraph, Google Gemini, Neo4j, LangChain, Python, Docker.

File Explorer Agent

2025

AI-powered file management assistant with a conversational Streamlit UI, LangChain orchestration, and 30+ MCP-based file tools.

- Built an AI-driven file management assistant with Streamlit, LangChain, and MCP server pattern.
- Designed multi-session chat UI with persistent history and async checkpointing using SQLite + asyncio.
- Integrated Azure OpenAI (AzureChatOpenAI) for LLM responses and 30+ MCP file tools for CRUD, search, and content analysis.
- Tech stack: Python, Streamlit, LangChain, Azure OpenAI, MCP, asyncio, aiosqlite, sqlite3.

Several other projects available on my GitHub: github.com/Rahulagowda004

ACTIVITIES & ACHIEVEMENTS

Hackathon Competitions

2023 – 2025

Demonstrated leadership and innovation across multiple hackathons

- Led the team to victory in VIT Hackathon, showcasing problem-solving and technical skills.
- Won Hackmania and Securathon, excelling in software innovation and cybersecurity.
- Participated in several other hackathons, sharpening rapid prototyping and AI integration abilities.

EDUCATION

B.E. in Artificial Intelligence and Machine Learning, Visvesvaraya Technological University

Bengaluru, Karnataka